Pranav Mishra

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Education

Master of Science, Computer Science ***Aug 2023 - May 2025***
University of Illinois at Chicago, Illinois, USA | [Graduate Assistant]
Coursework: Advanced ML, Applied AI, Advanced NLP, Computer Vision, VR, Game Design, Object-Oriented Programming

Bachelor of Science, Computer Science and Engineering ***Aug 2019 - June 2023***
Dayananda Sagar College of Engineering, Bangalore, Karnataka, India
Coursework: AI & ML, Software Engineering, Computer Networks, Operating Systems, Data Structures & Algorithms, DBMS

Technical Experience

Founding LLM Engineer | *alfred_ai* ↗ | New York City, NY ***April 2026 - Present***
- Building production-grade multi-agent LLM workflows for a consumer AI assistant serving 2K+ users; TypeScript backend infrastructure, orchestration reliability, tool-calling stability, and low-latency context retrieval across real-time pipelines

AI Engineer | *WheelPrice* ↗ | Charlotte, NC ***July 2025 - Mar 2026***
- Built React/Node.js/MongoDB CMS blog subsystem from scratch serving 10-20k daily users; implemented Redis caching, SSR, and dynamic SEO pipeline; reduced API response time 1.5s to 300ms via query optimization and connection pooling.
- Built internal analytics dashboard with XGBoost-based churn prediction and event-driven behavior tracking across sessions

Research Software Engineer | *UIC- VARE Lab* ↗ | Chicago, IL ***Feb 2024 - Present***
- Built RESTful APIs and PostgreSQL backend handling 1M+ records with auth middleware; deployed audio ML inference API at 150ms p95 latency via INT8 quantization; contributing to applied AI research in healthcare at UIC -TeamMedAgents

Research & Publications

TeamMedAgents: SLM-based multi-agent medical reasoning [ICML-2026 Submission] ***GitHub | Paper | SLMs-v2.0***
- Built multi-agent medical reasoning system with Google ADK achieving 77.63% accuracy across 8 benchmarks; Added tool-calling for external retrieval, reasoning traces for decision transparency, and trust-weighted voting for consensus
- Optimized 4B SLMs in multi-agent system; achieved 77.63% accuracy and 3.1× inference speedup vs frontier LLMs

MetaRAG: Metadata Enrichment for RAG Systems [ACCEPTED for IEEE CAI-2026] ***GitHub| IEEE Paper***
- Designed retrieval system achieving 82.5% precision and 0.925 Hit@10 through LLM metadata generation, cross-encoder reranking, and TF-IDF weighted embeddings; reduced hallucinations 25% via automated evaluation framework
- Deployed production LLM system on AWS using SageMaker and MLflow for model versioning and experiment tracking; handled 10k queries/day for code translation and automation tasks; maintained p99 latency <300ms with CI/CD pipelines

Projects

MockFlow-AI: Real-Time Voice Interview Platform with Multi-Agent Architecture ***GitHub | Website***
- Architected real-time voice interview platform with FSM-driven multi-agent orchestration with streaming STT-TTS pipelines
- Achieved sub-400ms end-to-end latency (5× faster than polling-based baseline); optimized WebSocket connections, implemented automated feedback loop, RAG for personalised questions, JWT authentication, and BYOK architecture

SnakeAI-MLOps: Multi-Agent Reinforcement Learning Snake Game ***GitHub | Demo***
- Engineered RL framework with reward shaping, experience replay & hyperparameter optimization in C++ with LibTorch
- Built MLOps pipeline with model versioning, automated testing, Docker CI/CD, and deployment monitoring with gameplay

MedRAG Avatar Platform - Intelligent Healthcare Conversational AI System ***GitHub | Demo***
- Built multi-modal RAG platform with FastAPI backend, Cosmos DB & Azure Speech Services achieving near 100% retrieval accuracy; implemented TTS generation, avatar rendering, and gesture animation with <250ms end-to-end latency

Skills & Extracurricular

TECHNICAL SKILLS: C#, C++, JavaScript, TypeScript, Python, LangChain, Java, React, ASP.NET Core, Node.js, FastAPI, Flask, SQL, PostgreSQL, MongoDB, NoSQL, Redis, Docker, Git, CI/CD, Azure, AWS, PyTorch, TensorFlow,

APPLICATIONS: Full-Stack Development, Front-End Development, Backend Development, Product-Led Growth, Customer-First Design, Data Analysis, Customer Insights, Collaboration with PMs & Designers, Scalable Software, Azure Cloud

Winner of MIT XR Hackathon | Built Meta Quest 3 app using Unity and Hugging Face for spatial data visualization ↗

INFORMS Analytics+ Speaker | Presented MetaRAG to 700+ professionals | First place HINT 5.0 Web3 museum for NFTs↗